Ram Calculus for Outer-Electron Radii of Neutral Coulomb Atoms: Boundary-Aware Mellin Profiles and Quantitative Tail-to-Radius Certificates

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M530

October 8, 2026

Abstract

We examine the OpenAI manuscript on generalized outer-electron radii for neutral, correlated, two-spin Coulomb atoms. Its target is an iterated large-charge and large-exterior-mass asymptotic, not convergence at a fixed number of outer electrons. Ram Calculus is applicable to the annular Poisson equation, its scale-invariant limiting profile, and the final inversion of an expected electron tail. We give exact boundary-aware Green and logarithmic Mellin representations, including angular modes, and prove finite-annulus stability and nonlinear reconstruction results. We derive a quantitative chain from conditional profile errors, packet widths, exceptional-event probabilities, and a separate far-tail bound to an explicit interval for the physical radius. A strengthened residual-based rigidity criterion and a spin-normalization extension are proved under stated hypotheses. These results improve error accounting and make several analytic interfaces testable; they do not replace the manuscript's many-body screening and intermediate-barrier theorems, independently certify its full quantum proof, or establish a finite-charge convergence rate. Contribution assessment: C3/5, notable source-specific quantitative certificates and constructive annular extensions, without an independently established improvement of the global quantum asymptotic.

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1 Problem, source status, and contribution boundary

The source is *Generalized outer-electron radii of neutral Coulomb atoms*, September 24, 2026, identified as M530 in the initial 722-manuscript screening catalogue [1]. The local study uses the repository snapshot `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The public repository and its family-263 formalization description were consulted on October 8, 2026. The current repository describes a collection at mixed verification stages; its present manuscript count need not equal the initial catalogue count. We preserve the original M530 identifier.

The family-263 documentation [3] reports a formalization of the iterated radius assertion and links to `CoulombRadii.lean`. The comparator defines the antisymmetric form domain, weak gradients, ground states, density, expected exterior mass, and the two ordered limits. It is a *challenge statement*, with a placeholder proof, not the submitted proof artifact. We do not mistake that placeholder for a defect in a separate implementation, and we have not run a Lean kernel build of that implementation. Our mathematical certification is restricted to the results proved here and the source modules explicitly assessed.

Applicability L4 and contribution C3 answer different questions. L4 means the source has very relevant analytic steps. C3 credits the error-to-radius certificate and finite-annulus analysis below. Classical Green identities, Mellin inversion, Sobolev theory, and maximum principles remain classical mathematics when expressed in Ram coordinates; no claim of historical priority or measured computational acceleration is made.

1.1 Exact many-electron model

For integer $Z \geq 1$, the nuclear charge and number of electrons both equal Z. On the antisymmetric spin space

$$\mathcal{H}_Z = \bigwedge^Z L^2(\mathbb{R}^3; \mathbb{C}^2)$$

the Hamiltonian is the self-adjoint operator associated with the semibounded Coulomb quadratic form on antisymmetric H^1 wavefunctions:

$$H_Z = \sum_{i=1}^{Z} \left(-\tfrac{1}{2}\Delta_{x_i} - \frac{Z}{|x_i|} \right) + \sum_{i<j} \frac{1}{|x_i - x_j|}. \tag{1}$$

There is no artificial spatial boundary and no magnetic gauge in this model. Coulomb singularities are handled by the form realization, not by assuming globally analytic wavefunctions. Let Ψ_Z be any normalized ground state and ρ_Z its spin-summed one-particle density. Then $\rho_Z \geq 0$, $\int \rho_Z = Z$, and

$$T_Z(r) = \int_{|x|>r} \rho_Z(x)\,\mathrm{d}x, \qquad R_m(\Psi_Z) = \inf\{r \geq 0 : T_Z(r) \leq m\}, \quad 1 \leq m < Z. \tag{2}$$

The density is integrable, so T_Z is continuous and nonincreasing, $T_Z(0) = Z$, and $T_Z(r) \to 0$. Continuity follows by dominated convergence because spheres have zero Lebesgue measure. The radius concerns expected electron mass, not a deterministic radius containing each electron, and not the radius of the ion with $Z - m$ electrons.

The source's main claim is, for every choice of the ground-state family,

$$\lim_{m\to\infty} m^{1/3} \limsup_{Z\to\infty,\, Z>m} R_m(\Psi_Z) = \lim_{m\to\infty} m^{1/3} \liminf_{Z\to\infty,\, Z>m} R_m(\Psi_Z) = \left(\frac{81\pi^2}{2}\right)^{1/3}. \tag{3}$$

All limits are integer limits. It does not assert a limit in Z for each fixed m, an error bound for a physical atom of specified charge, or a uniform simultaneous limit under only $Z/m \to \infty$.

1.2 Literature distinction

Solovej's Hartree–Fock result proves outer-radius control in an approximate variational model [4]; his later formulation distinguishes upper and lower large-charge limits in the generalized conjecture [5]. Thomas–Fermi–Dirac–von Weizsacker estimates concern another approximate model [6]. None can simply be substituted for a correlated Schrodinger ground state. The recent excess-charge estimate of Hundertmark, Pattakos, and Schulz concerns how many electrons an atom can bind, a different observable [7]. It is not a bound on the error in (3). The OpenAI foundation companion [2] supplies much stronger screening inputs for this particular source. Our use of those inputs remains explicitly conditional on them; an abstract or formalization description is not our independent proof of their full analytic derivation.

2 Source proof architecture and exact interfaces

Let $K(x) = |x|^{-1}$ and

$$k = \frac{2^{3/2}}{3\pi^2}, \quad a = 4\pi k, \quad A = \left(\frac{12}{a}\right)^2 = \frac{81\pi^2}{8}, \quad B = 4A. \tag{4}$$

The following interfaces keep quantum states, conditional fields, and limiting model fields distinct.

1. **Quantum estimates.** The source obtains a bounded energy offset from the foundation companion. For fixed $0 < \alpha < \beta$, its raw annular counts $N_{\alpha u,\beta u}$ satisfy

$$\|N_{\alpha u,\beta u}\|_{L^2(\mathbb{P})} \le C_{\alpha,\beta}(\max\{u^{-3},1\} + \sqrt{D_0 u}), \qquad \mathbb{E}N_{1,\infty} \le C(1+\sqrt{D_0}).$$

   The constants are uniform in the neutral ground state. This local L^2 bound, rather than the crude total count Z, removes exceptional data.

2. **Conditional smoothing.** Use radial unit-mass smooth packets $\mathcal{K}_x(y) = g_{t_x}(y-x)^2$, and the source's decreasing observation sigma-fields $\mathcal{F}_j$. Set

$$\mu_j = \mathbb{E}\left[\sum_i \mathcal{K}_{x_i} \mid \mathcal{F}_j\right], \qquad H_j = ZK - K * \mu_j.$$

   Every packet has mass one. Thus $\mu_j \ge 0$, $\int \mu_j = Z$, and $\Delta H_j = -4\pi Z\delta_0 + 4\pi\mu_j$ exactly. Conditional versions must be fixed before selecting a good datum.

3. **Inverse comparison and barriers.** On high-probability data the field has an upper Cr^{-4} cap and upper/lower implications between $r^6\mu_j$ and $r^4 H_j$. Separate intermediate subsolutions provide a positive lower profile bound and a nonnegative error measure of small expected mass. The same packet and observation families must serve both constructions. Changing the families while tightening accuracy would invalidate their combination.

4. **Intermediate scale.** Along $s_\ell \downarrow 0$ with $Z_\ell \ge Z_{\mathrm{bar}}(s_\ell)$ and $Z_\ell s_\ell^3 \to \infty$, choose retained data on expanding rescaled annuli. Set

$$F_\ell(x) = s_\ell^4 H_{j_\ell}(s_\ell x), \quad \sigma_\ell(x) = s_\ell^6 \mu_{j_\ell}(s_\ell x), \quad \lambda_\ell = Z_\ell s_\ell^3.$$

   Then $F_\ell = \lambda_\ell K - K * \sigma_\ell$, $\int \sigma_\ell = \lambda_\ell$, and $\Delta F_\ell = 4\pi\sigma_\ell$ away from zero. The total rescaled mass diverges, so global L^1 compactness is not available.

5. **Profile and count recovery.** Local compactness, positivity, and rigidity identify $F_*(x) = A|x|^{-4}$. Expected raw tails require, in addition, removing bad observations, removing packet smoothing, and a separate dyadic far-tail estimate. Radius inversion comes last.

2.1 A useful explicit inversion of the comparison

Write $v = r^4 H$ and $t = (r^6\mu/k)^{2/3}$. Suppose the source comparison is valid for heights $h \in [h_l, h_h]$, with cap $v \le C < h_h - \xi$. For $h_l < t < h_h$, its two implications give $|v_+ - t| \le \xi$ by approaching t from either side. For $t \le h_l$, both v_+ and t lie in $[0, h_l + \xi]$. The cap excludes $t \ge h_h$. Therefore

$$|r^6\mu - k(v_+)^{3/2}| \le \max\left\{\tfrac{3}{2}k\sqrt{h_h}\xi, k(h_l+\xi)^{3/2}\right\}. \tag{5}$$

This retains the possibly negative field in the low-density case; the comparison never gave an unconditional pointwise lower bound on H. Choose the heights for a desired accuracy first, then the small physical scale, then the charge threshold. A slow diagonal in the accuracy is valid without a rate for those thresholds, but it supplies no rate by itself.

3 Ram Calculus ingredients actually used

The private corpus [8, 9, 10] supplies a compatible representation and verification framework, not an extra hypothesis that all formal series converge. The following specific proof portions were used.

Supporting material	Role and retained restriction
PDE-first Astra TR8, elliptic weak operators, coercive inverse, and Green equivalence	A domain and trace are fixed before an inverse; a boundary lifting is retained; weak inverses are not automatically continuous pointwise kernels. The nonlinear monotonicity argument is adapted below.
Astra TR6, full Urysohn variation and Newton equation	All state-dependent terms are differentiated. The scalar nonlinearity here is a source-state term; an invented observation-state term would be as wrong as omitting a real one. Second derivatives require a positive corridor.
Ram Calculus Books, Volume 2, Chapter 1, Mellin inversion and energy	Logarithmic coordinates use $\mathrm{d}r/r$ and explicit weights. Endpoint terms are retained on finite annuli; a non-integrable homogeneous power is not assigned an ordinary full-line Mellin transform.

Each ingredient is rederived in the required setting below. The larger corpus was used as a routing index for pertinent PDE, kernel, and transform material, not treated as if every report had been independently proved and checked for this application. The local supporting files and read scope are identified in the accompanying evidence record.

4 Annular nonlinear Poisson problem and Green representation

Fix $0 < r_0 < R < \infty$ and the annulus $\Omega = \{r_0 < |x| < R\}$. For nonnegative boundary trace $b \in H^{1/2}(\partial\Omega)$, with a nonnegative H^1 lifting, consider

$$-\Delta F + a(F_+)^{3/2} = 0, \qquad \gamma F = b. \tag{6}$$

Let $V = H^1_0(\Omega)$, $\|v\|_V = \|\nabla v\|_2$, and let $G_D : V' \to V$ be the inverse of the weak Dirichlet $-\Delta$. The source class is well-defined: $H^1(\Omega) \hookrightarrow L^6(\Omega)$ in three dimensions, so $(F_+)^{3/2} \in L^4(\Omega) \subset V'$. If ℓ_b is any lifting,

$$F = \ell_b - G_D[-\Delta\ell_b + a(F_+)^{3/2}]. \tag{7}$$

The bracket is a weak source; the sign follows by setting $F = \ell_b + v$ in (6). On regular data the Green kernel can be localized by $y = x - u$ with the transformed integration domain $x - \Omega$. It is the Dirichlet annular kernel, not the free-space Coulomb kernel.

Proposition 4.1 (Existence, uniqueness, and residual stability)**.** *Equation (6) has one weak H^1 solution, and it is nonnegative. If $\widehat{F} \in H^1(\Omega)$ has the same trace and residual $\mathcal{R} = -\Delta\widehat{F} + a(\widehat{F}_+)^{3/2} \in V'$, then*

$$\left\|\nabla(\widehat{F} - F)\right\|_2 \le \|\mathcal{R}\|_{V'}. \tag{8}$$

No smallness assumption on a or on the solution is required.

Proof. Minimize $\frac{1}{2}\int|\nabla F|^2 + \frac{2a}{5}\int(F_+)^{5/2}$ over the affine trace space. Poincare's inequality for $F - \ell_b$ bounds minimizing sequences in H^1; convexity and weak lower semicontinuity give a minimizer.

The $H^1 \to L^{5/2}$ embedding justifies differentiation in every V direction. Testing by the negative part gives $F \geq 0$. The scalar map $t \mapsto (t_+)^{3/2}$ is monotone. Subtracting two equations and testing by their difference proves uniqueness. Subtracting the equation from the residual identity instead yields

$$\|\nabla e\|_2^2 + a \int [(\widehat{F}_+)^{3/2} - (F_+)^{3/2}]e = \langle \mathcal{R}, e \rangle, \quad e = \widehat{F} - F.$$

The second term is nonnegative and duality proves (8). □

This is an annular PDE result, not the existence theorem for a many-body ground state. A residual from only finitely many tests is not its full V' norm. Likewise (8) does not imply a pointwise error in three dimensions. Boundary mismatches must first be lifted and their contributions included in the residual.

5 Logarithmic master coordinates without assuming radiality

Put $r = e^t$, $\omega = x/|x|$, and

$$F(r, \omega) = Ar^{-4} v(t, \omega). \tag{9}$$

Using $r\partial_r = \partial_t$ and the spherical Laplacian gives the exact identity

$$\Delta F = Ar^{-6}(v_{tt} - 7v_t + \Delta_{\mathbb{S}^2} v + 12v).$$

Since $a\sqrt{A} = 12$, the profile equation is

$$P(v) := -v_{tt} + 7v_t - \Delta_{\mathbb{S}^2} v - 12v + 12(v_+)^{3/2} = 0. \tag{10}$$

The angular operator remains until a rigidity theorem eliminates angular dependence. Spherical averaging the nonlinearity first would be incorrect: $\overline{v^{3/2}}$ generally differs from $\bar{v}^{3/2}$.

For $v = 1 + w$, define $N(w) = (1 + w)_+^{3/2} - 1 - \frac{3}{2}w$. Then

$$Lw = -12N(w), \quad L = -\partial_t^2 + 7\partial_t - \Delta_{\mathbb{S}^2} + 6. \tag{11}$$

On $I = (t_0, t_1)$ with Dirichlet endpoints, its physical weak measure is $e^{-7t}\,\mathrm{d}t\,\mathrm{d}\omega$ and its form is

$$\int_{I \times \mathbb{S}^2} e^{-7t}(|w_t|^2 + |\nabla_\omega w|^2 + 6|w|^2)\,\mathrm{d}t\,\mathrm{d}\omega.$$

Writing $w = e^{7t/2} y$ conjugates the operator to

$$Lw = e^{7t/2}\left(-y_{tt} - \Delta_{\mathbb{S}^2} y + \frac{73}{4} y\right). \tag{12}$$

This is an exact weighted isometry, not a statement that unweighted Euclidean coefficient norms represent physical energy.

5.1 Angular Green blocks and endpoint liftings

Let $-\Delta_{\mathbb{S}^2} Y_{\ell j} = \ell(\ell + 1) Y_{\ell j}$ in an orthonormal spherical harmonic basis. For each mode put $\kappa_\ell = \sqrt{73/4 + \ell(\ell + 1)}$. The inverse kernel of the corresponding L block, integrated against $\mathrm{d}u$, is

$$K_\ell(t, u) = e^{7(t-u)/2} \frac{\sinh(\kappa_\ell(t_< - t_0)) \sinh(\kappa_\ell(t_1 - t_>))}{\kappa_\ell \sinh(\kappa_\ell(t_1 - t_0))}, \tag{13}$$

where $t_< = \min(t,u)$ and $t_> = \max(t,u)$. It vanishes at the two endpoints; its first-derivative jump at $t = u$ is -1. These facts and the homogeneous equation prove the inverse property. Nonzero endpoint data add the homogeneous lift, obtained for y from the two hyperbolic-sine boundary interpolants and then multiplied by $e^{7t/2}$. Weighted symmetry is $e^{-7t}K_\ell(t,u) = e^{-7u}K_\ell(u,t)$.

The homogeneous w exponents are

$$\lambda_{\ell,\pm} = \frac{7 \pm \sqrt{73 + 4\ell(\ell+1)}}{2}. \tag{14}$$

There is a growing and a decaying mode, including in the radial block. Simply discarding the growing coefficient changes a finite annular boundary-value problem. The classical radial decay exponent $(\sqrt{73} - 7)/2$ appears directly, but an indicial root alone does not prove a nonlinear asymptotic rate for the quantum density.

5.2 Finite Mellin endpoints and panels

For $\mathcal{M}_I h(s) = \int_{e^{t_0}}^{e^{t_1}} h(r) r^{s-1}\,\mathrm{d}r$, integration by parts gives

$$\mathcal{M}_I(rh')(s) = [r^s h(r)]_{e^{t_0}}^{e^{t_1}} - s\mathcal{M}_I h(s). \tag{15}$$

No endpoints may be dropped merely because the interval is described as a Ram window. On the full line, with $e^{ct}h(e^t) \in L^1 \cap L^2$, Fourier inversion yields

$$\int_0^\infty |h(r)|^2 r^{2c-1}\,\mathrm{d}r = \frac{1}{2\pi}\int_{\mathbb{R}} |\mathcal{M}h(c+i\xi)|^2\,\mathrm{d}\xi.$$

The pure profile Ar^{-4} fails full-line weighted integrability for every c, at one endpoint or the other. We therefore use finite annuli, or transform a localized remainder with its window commutator retained. If $\chi = \chi(t)$, then

$$[L,\chi]w = -2\chi' w_t - \chi'' w + 7\chi' w. \tag{16}$$

Piecewise polynomial Ram panels must preserve value and flux continuity or include interface distributions. Galerkin source coefficients require the mass matrix: the solution map is $K_h^{-1}M_h$, not K_h^{-1} on raw source coefficients. A finite matrix inverse by itself proves neither continuum convergence nor tail control.

6 Constructive annular extensions

Theorem 6.1 (A boundary-controlled nonlinear reconstruction)**.** *Let $I \times \mathbb{S}^2$ be a finite cylinder with prescribed endpoint values $b \in C(\mathbb{S}^2) \cap H^{1/2}(\mathbb{S}^2)$ on each end. Work in $X = C(\overline{I} \times \mathbb{S}^2)$ with its supremum norm. Let H_b be the homogeneous L-solution with those traces and $b_* = \max_{\partial I \times \mathbb{S}^2} |b|$. Set $\rho = 1/4$. If*

$$b_* + \frac{3\rho^2}{4\sqrt{1-\rho}} \le \rho, \tag{17}$$

then $w \mapsto H_b - 12L_D^{-1}N(w)$ is a contraction of the closed sup-norm ball $\|w\|_\infty \le \rho$. Its contraction constant is at most $q = 3(1-\sqrt{1-\rho}) < 1$. There is one solution in that ball. For an approximation $\widehat{w}$ in the ball its fixed-point defect gives

$$\|w - \widehat{w}\|_\infty \le \frac{\left\|\widehat{w} - H_b + 12L_D^{-1}N(\widehat{w})\right\|_\infty}{1-q}. \tag{18}$$

Proof. The maximum principle gives $\left\|L_D^{-1}\right\|_{\infty\to\infty} \le 1/6$ by comparison with the constant $\|f\|_\infty/6$, and $\|H_b\|_\infty \le b_*$. For $|w| \le \rho$, Taylor's theorem gives $|N(w)| \le 3w^2/(8\sqrt{1-\rho})$ and $|N'(w)| \le \frac{3}{2}(1-\sqrt{1-\rho})$. These imply invariance and the claimed contraction factor. The fixed-point theorem and its defect estimate prove the result. The harmonic lifting has finite energy by the $H^{1/2}$ trace condition, and the zero-boundary inverse of a continuous source lies in H_0^1. Thus the continuous fixed point is also a global weak H^1 solution. Elliptic regularity supplies classical smoothness for smooth endpoint data. The proof is uniform in cylinder length, although raw coordinate matrices can still be poorly conditioned. □

Theorem 6.2 (Pointwise corridor residual estimate)**.** *Let $v, \widehat{v} \in C^2(I\times\mathbb{S}^2)\cap C(\overline{I}\times\mathbb{S}^2)$ satisfy $v, \widehat{v} \ge m_0 > 4/9$. If $P(v) = 0$, $\|P(\widehat{v})\|_\infty \le \epsilon$, and their endpoint discrepancy is at most d, then*

$$\|v-\widehat{v}\|_\infty \le \max\{d, \epsilon/(18\sqrt{m_0}-12)\}. \tag{19}$$

Proof. Their difference satisfies a linear drift-elliptic equation whose zero order coefficient is $c = -12+18\int_0^1 \sqrt{v+\theta(\widehat{v}-v)}\,\mathrm{d}\theta \ge 18\sqrt{m_0}-12 > 0$. Comparing with the positive and negative constant on the right of (19) proves the result. □

The positive corridor is a real restriction of this logarithmic sup-norm certificate. It is not needed for the physical-coordinate weak estimate (8). Replacing a verified continuum residual by a residual at finitely many sample points would not satisfy the theorem.

7 Quantitative tails and radius inversion

The decisive observable is a tail, not a local field norm. This section provides explicit bounds that can accept valid source estimates without pretending to compute their currently unspecified constants.

Theorem 7.1 (Finite-annulus density-to-tail bound)**.** *Let $\sigma \ge 0$ be locally integrable outside zero, with finite tail $T_\sigma(r)$. Suppose on $r \le |x| \le R$,*

$$\sigma(x) = kA^{3/2}|x|^{-6}\{v(\log|x|,\omega)_+^{3/2} + e(x)\}, \quad |v-1| \le \delta < 1, \quad |e| \le \eta,$$

and $T_\sigma(R) \le DR^{-3}$. Write $\vartheta = (1+\delta)^{3/2} - 1 + \eta$. Then

$$|T_\sigma(r) - Br^{-3}| \le B\vartheta(r^{-3}-R^{-3}) + (D+B)R^{-3}. \tag{20}$$

Proof. Both deviations of $(1+w)^{3/2}$ from one, for $|w| \le \delta$, are bounded by $(1+\delta)^{3/2}-1$. Integrating $|x|^{-6}$ over the annulus gives $4\pi(r^{-3}-R^{-3})/3$. The constants obey $4\pi kA^{3/2}/3 = 4A = B$. Split the tail at R and use positivity and the stated far-tail bound for the remaining difference. □

The outer term cannot be removed on the basis of local convergence alone. For example a nonnegative mass packet escaping to radius $R_n \to \infty$ vanishes on every fixed annulus yet changes all tails below R_n by its mass. A local PDE approximation needs a separate exclusion of this loss of mass at infinity.

Theorem 7.2 (Two-sided radius certificate)**.** *Let T be a continuous nonincreasing nonnegative tail on $[0,\infty)$, tending to zero, and let $R_m = \inf\{r \ge 0 : T(r) \le m\}$. Suppose on $[a_0, b_0]$, with $B_\pm > 0$ and $0 \le e < m$,*

$$B_- r^{-3} - e \le T(r) \le B_+ r^{-3} + e. \tag{21}$$

Define $r_- = (B_-/(m+e))^{1/3}$ *and* $r_+ = (B_+/(m-e))^{1/3}$. *If* $0 < a_0 < r_- \le r_+ \le b_0$, *then*

$$r_- \le R_m \le r_+. \tag{22}$$

In particular, if $B_\pm = B(1 \pm \delta)$, $0 \le \delta < 1$, *and* $\zeta = e/m < 1$, *then*

$$\left(\frac{1-\delta}{1+\zeta}\right)^{1/3} \le \frac{R_m}{(B/m)^{1/3}} \le \left(\frac{1+\delta}{1-\zeta}\right)^{1/3}. \tag{23}$$

Proof. For $a_0 \le r < r_-$, the strict lower inequality gives $T(r) > m$. Monotonicity extends exclusion to $0 \le r < a_0$. At r_+ the upper bound is at most m. These statements prove the infimum bounds without strict monotonicity or differentiability of the tail. Division gives (23). The strict placement $a_0 < r_-$ prevents an uncontrolled plateau at level m before the certified interval. □

For small δ, ζ the first-order relative error is $(\delta + \zeta)/3$; the exact bounds, rather than that linearization, are the certificate. If $\delta = 0.03$, $\zeta = 0.01$, the factors are exactly $(97/101)^{1/3}$ and $(103/99)^{1/3}$. These illustrative error budgets are not claimed measurements on quantum atoms.

7.1 Exceptional observations with the right moment

Let $C_A = \int_A \sum_i \mathcal{K}_{x_i}$ be a smoothed raw annular count and $G \in \mathcal{F}_j$ have $\mathbb{P}(G^c) \le p$. If A^+ contains every center whose packet meets A, then $0 \le C_A \le N_{A^+}$. Thus

$$\mathbb{E}\left[\mathbf{1}_{G^c} \int_A \mu_j\right] = \mathbb{E}[\mathbf{1}_{G^c} C_A] \le \sqrt{p}\, \|N_{A^+}\|_2 \,. \tag{24}$$

The equality is the conditional-expectation identity using G-measurability; the inequality is Cauchy–Schwarz. Under rescaling, multiply by s^3. The source annular moment bound makes this $O(\sqrt{p})$ at a fixed rescaled annulus, whereas $s^3 Z\sqrt{p}$ need not vanish. On good data a profile error E_A contributes at most E_A to the expected annular mass error, where E_A means $\sup_G |\int_A \mu_j - \int_A \rho_*| \le E_A$ after any density error has been integrated. The missing fraction of model mass contributes at most $p \int_A \rho_*$. Consequently a valid combined bound is

$$\left|\mathbb{E}\int_A \mu_j - \int_A \rho_*\right| \le E_A + p\int_A \rho_* + \sqrt{p}\, \|N_{A^+}\|_2 \,, \tag{25}$$

with all terms in one chosen coordinate and mass normalization. This estimate needs no independence between the good event and the count.

7.2 Packet removal as a multiplicative radius budget

Suppose each packet at a center x in the relevant outer region is supported in $|y - x| \le \nu|x|$, $0 \le \nu < 1$. Suppose inner-core packets cannot reach any threshold used below. Taking expected counts gives

$$T_{\mathrm{sm}}((1+\nu)r) \le T_{\mathrm{raw}}(r) \le T_{\mathrm{sm}}((1-\nu)r). \tag{26}$$

Indeed, a packet lying beyond $(1+\nu)r$ can only originate from a center beyond r, and every center beyond r places all of its packet beyond $(1-\nu)r$. Integrate the packet indicators and use the tower identity. For the source widths

$$t_x = c_1 \widehat{d}_x \min\{\widehat{d}_x, s\}^w, \quad \widehat{d}_x = \max\{|x|, r_0\}, \quad w = 10^{-5},$$

one may take $\nu = c_1 s^w$ for $|x| \geq r_0$. Core packets lie inside $(1+\nu)r_0$; thresholds larger than that number exclude them. This is a geometric support bound, not a smoothness assumption on the raw density.

Applying Theorem 7.2 to the smoothed tail envelopes and (26) yields

$$\frac{1}{1+\nu}\left(\frac{B_-}{m+e}\right)^{1/3} \leq R_m^{\mathrm{raw}} \leq \frac{1}{1-\nu}\left(\frac{B_+}{m-e}\right)^{1/3}, \tag{27}$$

under the following explicit sufficient conditions:

$$\max\{a_0, (1+\nu)r_0\} < r_- \leq r_+ \leq b_0, \quad 0 \leq e < m, \quad B_\pm > 0, \quad 0 \leq \nu < 1. \tag{28}$$

Here $[a_0, b_0]$ is the certified smoothed-tail interval and $r_\pm$ are the unshifted smoothed roots. For raw radii just below $r_-/(1+\nu)$, the lower sandwich evaluates the smoothed tail just below r_- but above both a_0 and the core packet radius. It is strictly greater than m, and monotonicity excludes all earlier raw radii. At $r_+/(1-\nu)$, the upper sandwich evaluates exactly at r_+, where the smoothed tail is at most m. This proves (27) without requiring unused shifted radii to lie in the interval. This supplies a separate, explicit packet contribution to the radius error rather than absorbing it in an unspecified asymptotic remainder.

7.3 One complete composite certificate

To make the composition precise, fix a single coordinate normalization and $0 < a_0 < b_0 < R$. Suppose the good-data density assumptions of Theorem 7.1 hold on $a_0 \leq |x| \leq R$ with common δ, η, on an event of complement probability at most p. Suppose every smoothed raw annular count contributing to $\{r < |x| < R\}$, $a_0 \leq r \leq b_0$, is dominated by a common enlarged raw count of L^2 norm at most C_{bad}. Finally suppose the *unconditional* smoothed tail at R is at most DR^{-3}. Set

$$\delta_T = (1+\delta)^{3/2} - 1 + \eta + p, \qquad e_T = C_{\mathrm{bad}}\sqrt{p} + (D+B)R^{-3}. \tag{29}$$

Then uniformly on $[a_0, b_0]$ the unconditional smoothed tail satisfies (21) with $B_\pm = B(1 \pm \delta_T)$ and $e = e_T$. Indeed the good annular error is at most $B\vartheta r^{-3}$, the lost model fraction is at most pBr^{-3}, the bad contribution is at most $C_{\mathrm{bad}}\sqrt{p}$, and the remaining tail costs $(D+B)R^{-3}$. This proves the composite assertion directly by splitting the expected tail, rather than assuming that a conditional far-tail bound automatically controls exceptional observations.

The radius conclusion additionally requires $\delta_T < 1$, $e_T < m$, and the root-placement conditions of Theorem 7.2, or (28) for the raw radius. Merely having $\delta < 1$ is not enough for these conditions. A certified profile bound from Section 6 may supply δ, but its boundary and continuum-residual hypotheses must hold on the good data. If the source scaling $x = y/s$, $\sigma(x) = s^6\mu(sx)$ is used, the target mass in this certificate is $m_s = s^3 m_{\mathrm{physical}}$ and the physical radius is s times the certified radius. For $s = m_{\mathrm{physical}}^{-1/3}$ the target rescaled mass is one. Mixing the two mass units would give an incorrect certificate.

8 A quantitative all-scale rigidity extension

The source's exact rigidity argument is already nonradial. In logarithmic coordinates it admits the following residual extension. The lower bound in the statement is an input, not a consequence of nonnegative spherical means.

Theorem 8.1 (Residual rigidity on the whole logarithmic cylinder)**.** *Let $v \in C^2(\mathbb{R} \times \mathbb{S}^2)$ satisfy $0 < m_0 \le v \le M_0 < \infty$ and $|P(v)| \le \epsilon$ everywhere. Then*

$$\left(\max\left\{0, 1 - \frac{\epsilon}{12m_0}\right\}\right)^2 \le v(t,\omega) \le \left(1 + \frac{\epsilon}{12}\right)^2. \tag{30}$$

In particular $P(v) = 0$ implies $v \equiv 1$ and hence $F = Ar^{-4}$.

Proof. Put $g(z) = 12z(\sqrt{z}-1)$, $m = \inf v$, and $M = \sup v$. For $\eta > 0$, the function $v(t,\omega) - \eta\sqrt{1+t^2}$ attains its maximum. At a maximizer $|v_t| \le \eta$ and $v_{tt} + \Delta_{\mathbb{S}^2} v \le \eta$, so $P(v) \ge g(v) - 8\eta$. The values of v at such maximizers approach M as $\eta \downarrow 0$: compare with any fixed approximate maximizer of v, and then let its error decrease. Thus $g(M) \le \epsilon$. Minimizing $v + \eta\sqrt{1+t^2}$ similarly gives $g(m) \ge -\epsilon$. If $M \ge 1$, then $\sqrt{M} - 1 \le \epsilon/(12M) \le \epsilon/12$; if $M < 1$ the asserted upper bound already holds. If $m \le 1$, then $1 - \sqrt{m} \le \epsilon/(12m) \le \epsilon/(12m_0)$; if $m > 1$ the lower bound already holds. Squaring the nonnegative lower bound for $\sqrt{m}$ proves the theorem. No attainment of m or M, radial symmetry, or global integrability of F was used. □

For a physical residual $\Delta F - aF^{3/2}$, equation (9) shows that its natural bound here is $\sup r^6|\Delta F - aF^{3/2}|/A \le \epsilon$. An absolute residual bound in unweighted physical coordinates is a different assertion. The theorem is useful for an all-scale approximate profile, but a finite collection of panels cannot establish its global lower bound and global residual without additional tail certificates. In particular the zero solution would invalidate exact rigidity if $m_0 > 0$ were removed.

8.1 A spin-count normalization extension

For a model with q equally populated spin states and kinetic term $-\Delta/2$, the phase-space density coefficient is

$$k_q = \frac{q}{6\pi^2} 2^{3/2} = \frac{q\sqrt{2}}{3\pi^2}.$$

It follows by counting a momentum ball of radius $\sqrt{2F}$ with state density $q/(2\pi)^3$. Repeating the local Poisson balance yields

$$A_q = \frac{81\pi^2}{2q^2}, \qquad B_q = 4A_q = \frac{162\pi^2}{q^2}, \qquad R_m^{\text{model}} = \left(\frac{B_q}{m}\right)^{1/3}. \tag{31}$$

The normalized cylinder equation and its local certificates are unchanged. This is an extension of the *model equation*, conditional on that phase-space closure. It is not a proof of the quantum asymptotic for arbitrary q, bosons, relativistic kinetic energy, external magnetic fields, or molecules. Those changes require new foundation estimates; several would change the scaling itself.

9 How the source reaches the physical radius

The source profile and tail modules received a separate bounded mathematical audit. No verified error was found there conditional on the stated foundation inputs. The following details are essential rather than optional technicalities.

9.1 Signed compactness and positivity are different stages

For a finite neutral conditional field $F_\ell = \lambda_\ell K - K * \sigma_\ell$, spherical averaging gives

$$\overline{F}_\ell(d) = \int_{|z|>d} \left(\frac{1}{d} - \frac{1}{|z|}\right) \sigma_\ell(z)\,\mathrm{d}z \geq 0. \tag{32}$$

This is Newton's angular-average identity; it requires no radial density. Combined with the upper cap, it bounds the full spherical L^1 norm by $8\pi C d^{-4}$. On a compact ball away from zero, let $P_\ell = K * (\sigma_\ell \mathbf{1}_Q)$. Its value and first weak derivative are bounded because the Coulomb kernel and its first derivative are locally integrable in three dimensions and the local density is bounded. The sum $F_\ell + P_\ell$ is harmonic. Harmonic interior estimates then give local compactness of the signed fields, despite diverging total mass. The limiting equation is obtained against compactly supported smooth tests, not by differentiating a uniform limit.

The positive lower bound comes from the separate intermediate subsolution. In the source notation, subtracting the field equation from the whole-space subsolution inequality cancels the nuclear Dirac mass and the entire inner density before testing. A nonnegative Coulomb correction removes the small inner error and the comparison-band error. For $a_\ell \in [1/(2q_\ell), 1/q_\ell]$, the latter has mass $O(q_\ell^{-8} a_\ell^{-3}) = O(q_\ell^{-5})$. Small mass alone does not control its potential at a point; its extra local density bound near that point supplies the missing control. The positive part of the corrected difference is an admissible H_0^1 test. No derivative of a sharply truncated field is taken, so no hidden interface measure is discarded. The source then proves a positive r^{-4} lower bound and obtains the exact nonradial profile by dilation and contact. The zero-residual case of Theorem 8.1 gives an alternative classical formulation once local regularity and the two-sided global bound have been established.

9.2 Tightness, normalization, and ordered limits

The rescaled expected count is s^3 times the physical expected count, not s^6 times that count; s^6 is the density scaling before changing variables. The source sums shells of physical radii $2^p M s$ below one. Its local count estimate gives a geometric factor $\sum_{p\geq 0} 2^{-3p} = 8/7$. The independent count outside physical radius one contributes only $O(s^3)$. Hence

$$\limsup_{\ell\to\infty} s_\ell^3 T_{Z_\ell}(M s_\ell) \leq C M^{-3}.$$

First take the sequence limit on a fixed annulus, then let $M \to \infty$. This is exterior tightness away from the origin, not probability-measure tightness of a sequence with bounded total mass. Together with (24) and packet removal, it gives the source's limiting tail Br^{-3} at every fixed positive rescaled radius.

For the radius conclusion, one may choose a finite integer threshold

$$Z_{\min}(m) \geq \max\{m+1, Z_{\mathrm{bar}}(m^{-1/3}), m^{4/3}\}. \tag{33}$$

Every sequence $m \to \infty$, $Z \geq Z_{\min}(m)$ is admissible with $s = m^{-1/3}$. At fixed $a_- < B^{1/3} < a_+$ the limiting tail ratios are $B/a_-^3 > 1$ and $B/a_+^3 < 1$. If the corresponding radius bracket failed uniformly for arbitrarily large m, choose a violating charge and ground state along that sequence; the two strict limiting inequalities give a contradiction. This proves the stronger state-uniform conclusion

$$\sup_{Z\geq Z_{\min}(m),\, \Psi_Z \text{ ground state}} \left| m^{1/3} R_m(\Psi_Z) - B^{1/3} \right| \longrightarrow 0, \tag{34}$$

conditional on the source inputs. For each fixed m, its finite threshold removes only finitely many charges, so both inner large-Z limits inherit the bracket and (3) follows. This stronger uniform phrasing is already a consequence of the source's proof; it is not credited as an independent Ram discovery. No bound on the growth of Z_{bar} has been established here.

10 Implementable verification and its limits

A practical boundary-aware implementation can proceed as follows.

1. Specify the annulus, physical boundary traces, angular resolution, and whether the observable concerns a model density, conditional density, or actual expected raw count. Keep A, k, B and spin normalization explicit.

2. Use logarithmic radial panels and spherical harmonic modes. Form the physical Gram matrices, boundary lifts, and the Green blocks (13); retain angular coupling in the nonlinear term.

3. Use the full derivative of the nonlinearity. On a positive corridor, $D[(1+w)^{3/2}]h = \frac{3}{2}\sqrt{1+w}\,h$ and $D^2[(1+w)^{3/2}][h,k] = \frac{3}{4}(1+w)^{-1/2}hk$. The second derivative is not bounded through $w=-1$. Clipping there changes the differentiability class.

4. Certify boundary errors and a continuum residual or fixed-point defect, including panel truncation, angular truncation, quadrature, window commutators, and linear-solve error. Use Theorem 6.1 or 6.2 only when its hypotheses are satisfied. A small floating-point matrix residual alone is insufficient.

5. Convert the profile bound to density and annular mass. Add the exceptional-event term with a localized count moment, then a separate far-tail bound. Record packet width ν and the certified radial interval.

6. Apply (22) or (27). Reject an interval if $e \geq m$, if a shifted radius leaves the certified band, or if a core packet can cross a tested threshold.

The companion verification script tests the coefficient balance, cylinder conjugation, angular indicial roots, Green derivative jump, nonlinear constants, finite Mellin endpoint term, exact model tails, packet radius factors, and adversarial examples for missing endpoints and escaping mass. It is an algebraic and finite-test validation of formulas, not a numerical solution of a Z-electron Hamiltonian, a rigorous interval computation of an actual atom, a formal proof-assistant check, or a timing benchmark. The displayed analytic proofs are the justification of the continuum statements.

11 Pros, cons, and future directions

11.1 Advantages over the source presentation

The source establishes a qualitative asymptotic through compactness, rigidity, and uniform-integrability arguments. The present formulation adds an explicit division of errors by their physical role. The pointwise field certificate, density conversion, exceptional-observation estimate, far-tail bound, and packet-radius factor are individually inspectable. The exact inversion theorem works for non-strictly monotone tails and does not require differentiating a noisy density. The all-scale residual rigidity theorem quantifies the source's exact profile step if its global comparability assumptions are available.

Logarithmic coordinates display all angular channels and both indicial branches. The Green blocks retain endpoint data, and the transformed coercive form specifies the correct metric. The constructive contraction result gives a reproducible local model solve with a computable defect constant. These features can improve transparency, adaptation, and error control even when the original asymptotic coefficient does not change.

11.2 Costs and limitations

The many-body screening, offset, barrier propagation, and localized count theorems remain the difficult source inputs. No coordinate transform proves them. A positive lower corridor is needed for some of our strongest pointwise bounds; it is stronger than mere local convergence and cannot be assumed for every finite-system signed screened field. Full-line residual rigidity requires all-scale control, while practical panels cover only a finite interval. The small contraction ball is a restricted reconstruction theorem, not a universal nonlinear solver.

Angular resolution, boundary certification, and trustworthy far-tail constants have a cost. Weighted conjugation improves analytic structure but may amplify raw coefficients over a long logarithmic interval. The conditional error chain is only as effective as the constants provided to it. There is no demonstrated speedup, smaller physical radius coefficient, improved quantum convergence rate, or independently completed proof of the foundation companion.

11.3 Research that would turn the certificates into effective bounds

The highest-priority next step is to make the comparison thresholds and $Z_{\mathrm{bar}}(s)$ effective, retaining their dependence on annular width, accuracy, and propagation parameters. A second step is an interval-verified annular implementation with certified boundary data, angular truncation, and flux residuals. Matching adjacent logarithmic annuli could then reduce the large global lower-bound assumption to checked local corridors plus an independently established tail argument.

Sharper far-tail constants and improved localized moments would directly improve (27). The square-root exceptional-probability loss could be replaced by $p^{1-1/q}$ if a valid localized L^q count bound with $q > 2$ were proved, by Holder's inequality; such a bound is not supplied here. Exploring q-spin or molecular analogues requires checking the quantum-to-model passage anew, despite the simple model normalization (31). Finally, formalizing the residual and radius certificates would complement, rather than substitute for, the upstream quantum proof artifacts.

12 Assessment and conclusion

The demonstrated contribution is **C3/5**. The specific additions are Theorems 6.1, 6.2, 7.1, 7.2, and 8.1, together with the explicit exceptional-event and packet budget leading to (27). They are proved under displayed hypotheses and concern actual steps in the source, not a replacement toy problem. Some constituent estimates are standard; the credited contribution is their source-specific combination and quantitative extension. No C4 or C5 credit is assigned for inherited source conclusions or for future work.

Ram Calculus is particularly useful here as a disciplined way to connect scale-adapted representations with boundary conditions and observable errors. It leaves the source coefficient $B^{1/3}$ unchanged and identifies exactly what additional quantitative input is needed for an effective finite-charge radius theorem. This supports further investigation while keeping the boundary between a proved local extension and an unverified global quantum claim clear.

A Read scope and evidence boundary

The parent read source Sections 1–3 and the foundation-interface appendix, the constants and bibliography, and the complete independent audit of Sections 4–5. The independent reader examined Sections 1–5 in full and the cited nonradial Sommerfeld lemma in Solovej's publisher PDF. Those downstream arguments were checked conditional on their displayed inputs. The foundation companion's complete proof was not independently replayed. The locally read Ram proof portions are the elliptic, Green, and semilinear sections of Astra TR8; full fixed-domain variation in Astra TR6; and Volume 2, Chapter 1's Mellin section. Reading an index or a title is not counted as reading a proof. Source hashes, independent findings, test results, and PDF checks accompany the report locally. The independent review of the present local theorems prompted the explicit $H^{1/2}$ boundary condition, core-packet inequalities, and uniform composite budget above. It found no counterexample to the stated operator, residual-rigidity, or spin-normalization identities. The executed suite contains 236 passing symbolic and finite checks; these do not certify any particular quantum state or replace the analytic proofs.

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